Bayesian SC vs Bayesian SDID

Geo-Lift Comparison on Simulated Panel

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June 18, 2026

1 What this project does

Company S uses Bayesian Synthetic Control (BSC) for geo-lift today and is in the process of upgrading to Bayesian Synthetic Difference-in-Differences (BSDID). This project fits both models on the same simulated panel with a known treatment effect and reports the difference.

2 Data

A 21-unit, 40-week panel. One treated unit, 20 controls. The treatment hits the treated unit at week 31 with a true per-week effect of $\tau = 25{,}000$ SEK. Each unit has its own baseline, a shared trend, annual seasonality, and Gaussian noise.

3 The two models

BSC models the pre-period as a weighted average of controls and extends the same weights into the post-period:

$$Y_t^{(0)} \sim \mathcal{N}\left(\sum_i \omega_i Y_{i,t}^{(C)},\ \sigma^2\right), \quad \omega \sim \text{Dirichlet}(1).$$

The ATT is the post-period residual $Y_t^{(0)} - \sum_i \omega_i Y_{i,t}^{(C)}$.

BSDID adds a second simplex of time weights λ on the pre-period and subtracts the λ-weighted pre-period gap from the ATT:

$$\text{ATT}_t = \left(Y_t^{(0)} - \omega \cdot Y_t^{(C)}\right) \ - \ \sum_s \lambda_s \left(Y_s^{(0)} - \omega \cdot Y_s^{(C)}\right).$$

λ has a Dirichlet prior and is anchored to data through a likelihood that requires the λ-weighted pre-period of each control to predict that control's post-period level:

$$\overline{Y_{i,\text{post}}^{(C)}} \sim \mathcal{N}\left(\sum_t \lambda_t\, Y_{i,t}^{(C)},\ \sigma_\lambda^2\right), \quad i = 1, \ldots, N.$$

The observation noise has a weakly informative Half-Normal prior anchored to the empirical pre-period standard deviation, $\sigma \sim \text{HalfNormal}(0, \widehat{\sigma}_{\text{emp}})$.

4 Inference

Both models fitted with Stan via cmdstanpy, four chains of 1 500 warmup + 1 500 sampling iterations each. All convergence diagnostics passed: no divergences, $\widehat{R} \approx 1$, ESS satisfactory.

5 Results

	Truth	**BSC**	**BSDID**
Mean ATT	25 000	19 550	29 333
90% interval	–	[14 386, 23 889]	[26 154, 32 301]
Interval width	–	9 503	**6 147**
|Error|	–	5 450	**4 333**

Table 1: ATT posterior summary on one simulated panel. BSDID has both a smaller absolute error and a 35% tighter interval than BSC. A Monte Carlo study over many panels would be required to separate systematic bias from sampling noise.

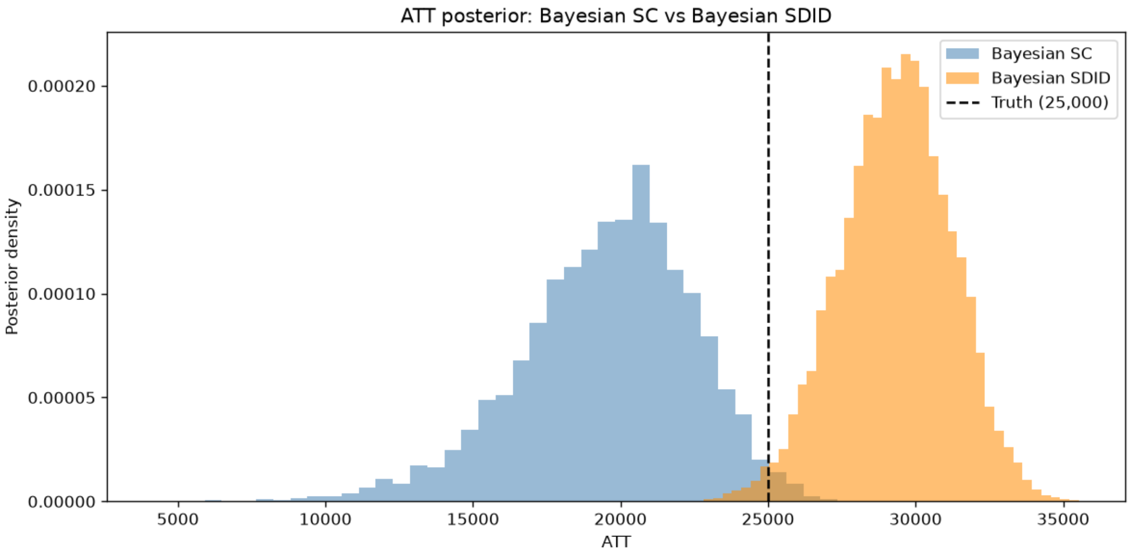

Figure 1: ATT posteriors. Dashed line is truth.

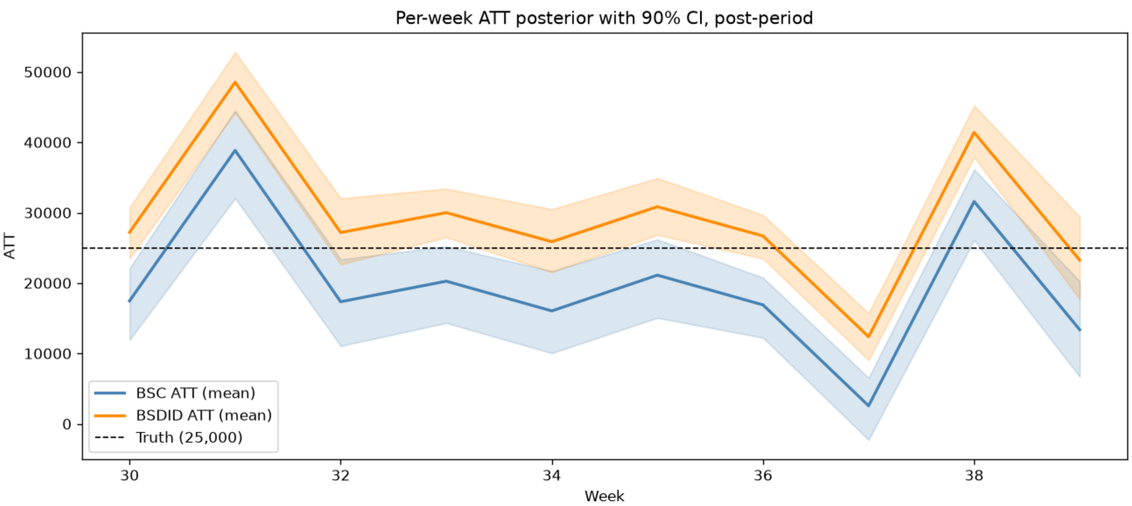

Figure 2: Per-week ATT, posterior mean and 90 % CI.

6 Why BSDID is expected to win

The treated unit sits slightly below its synthetic control in the pre-period. BSC reads that offset as part of the post-period treatment effect and underestimates τ by 22 %. BSDID subtracts the offset out of the ATT formula. On this panel it overshoots by 17 %, but with a tighter posterior and a smaller absolute error.

This is consistent with the structural improvement Arkhangelsky et al. predict. It matters whenever the treated region happens to be a little above or below the convex hull of its controls in the pre-period, which is the common case on real geo data.

7 Limitations

Single seed. Uniform Dirichlet priors. No smoothness regularization on λ beyond the Dirichlet. Pure on/off treatment. Production-grade Company S would add informative priors and a hierarchical structure pooling across past experiments.